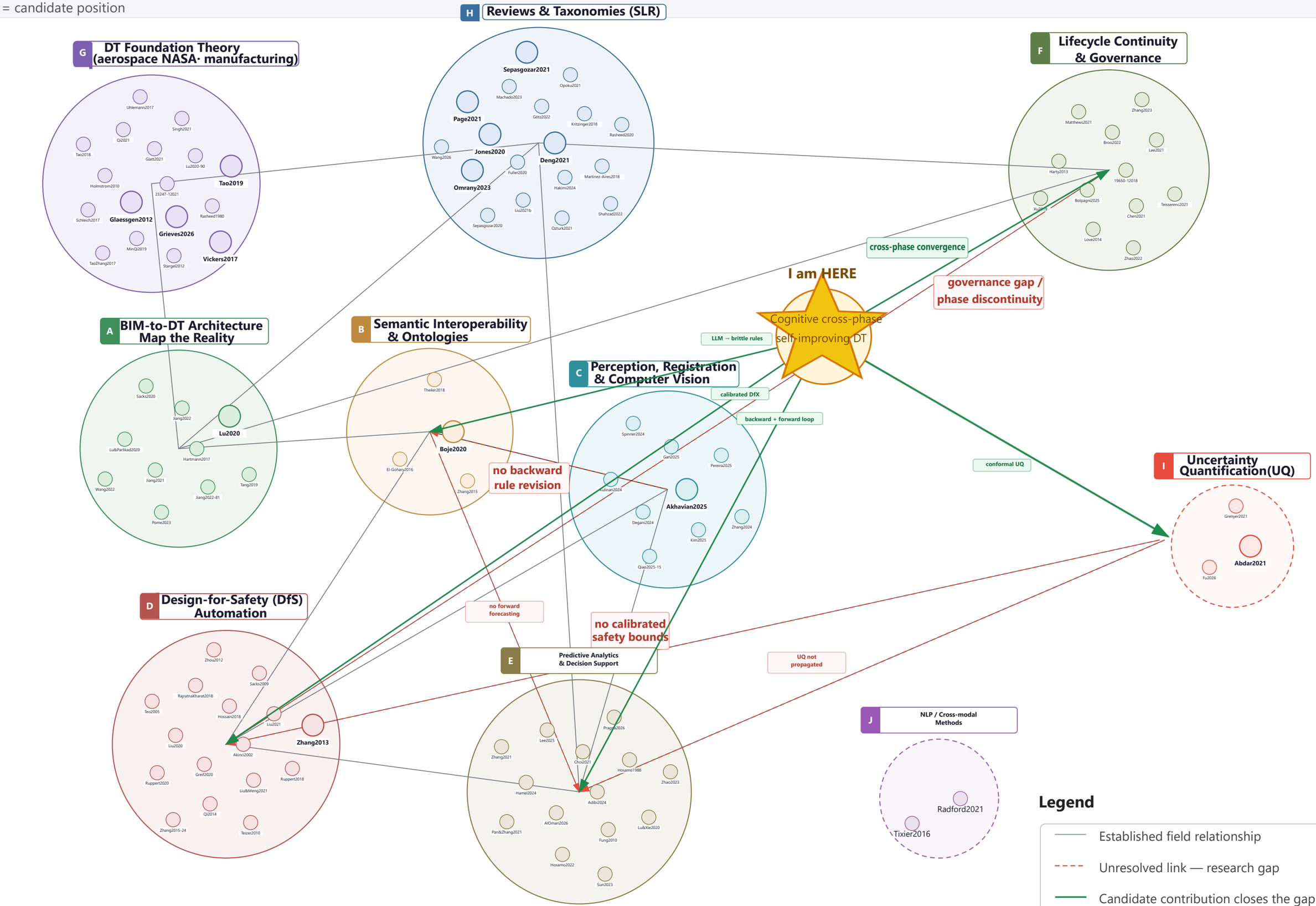
**AECO Digital-Twin Research Landscape — 103 papers**

Solid lines = established field relationships ·
Dashed red = unresolved gaps ·
★ = candidate position




## Legend

- Established field relationship
- Unresolved link — research gap
- Candidate contribution closes the gap
- Larger bubble = higher influence paper